

3B2 FPGA Programming Lab Report

Liu Zihe^{1*}

¹University of Cambridge

Abstract

This report documents the design, implementation and testing of a traffic light controller on an Altera Cyclone V FPGA. A three-state finite state machine controlling a pedestrian crossing is described in Verilog, compiled, and configured onto a DE1-SoC board via JTAG, and is improved with a vehicle lockout.

1 Traffic Light Controller Design

We design a controller for a pedestrian light-controlled crossing, satisfying the following requirements:

1. The initial state is vehicle GREEN and pedestrian RED.
2. State changes occur only on demand: a pedestrian request or a reset.
3. After a request, the sequence is: vehicle YELLOW for 5 s, then vehicle RED with pedestrian GREEN for 10 s, then back to the initial state.
4. An asynchronous reset returns to the initial state from any state.

These requirements are captured by a three-state FSM shown in Figure ?? . State G is the idle state with vehicle GREEN and pedestrian RED. A pedestrian request triggers a transition to state Y (vehicle YELLOW), which serves as a warning to vehicles. After a 5 s timeout the FSM enters state R (vehicle RED, pedestrian GREEN), allowing pedestrians to cross. A 10 s timeout then returns the FSM to state G. An asynchronous active-low reset forces the FSM back to state G from any state.

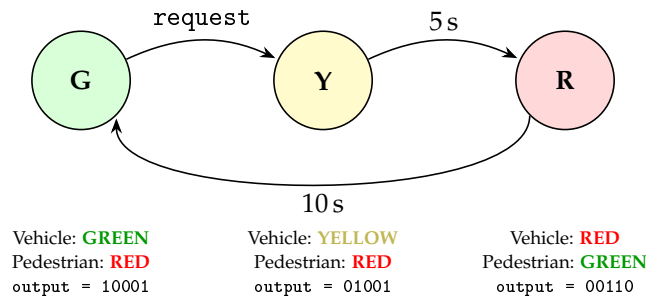


Figure 1: State diagram for the basic TLC FSM. The five outputs are grouped into a single vector $\text{output}[4:0] = \{\text{veh_G}, \text{veh_Y}, \text{veh_R}, \text{ped_G}, \text{ped_R}\}$. reset (active LOW, asynchronous) returns to state G from any state.

The outputs of each state and the transitions to the next states are summarised in Table ?? . Each state activates exactly one vehicle light and one pedestrian light.

Table 1: State table for the traffic light controller.

State	Next State				Outputs				
	reset	request	5 s	10 s	veh_G	veh_Y	veh_R	ped_G	ped_R
G	G	Y	–	–	1	0	0	0	1
Y	G	–	R	–	0	1	0	0	1
R	G	–	–	G	0	0	1	1	0

*CRSID: zl559. Date: 20th February 2026

2 Verilog Implementation

Listing 1: Basic traffic light controller.

```
1 module tlc (
2     input wire      clk,
3     input wire      request,
4     input wire      reset,
5     output reg [4:0] \output // {veh_G, veh_Y, veh_R, ped_G, ped_R}
6 );
7
8     localparam G = 2'd0, Y = 2'd1, R = 2'd2;
9
10    reg [1:0] state;
11    reg [28:0] count;
12
13    // Sequential: state transitions & timer
14    always @(posedge clk or negedge reset) begin
15        if (!reset) begin
16            state <= G;
17            count <= 29'd0;
18        end else begin
19            case (state)
20                G: begin
21                    if (request == 1'b0) begin
22                        state <= Y;
23                        count <= 29'd0;
24                    end
25                end
26                Y: begin
27                    if (count == 29'd250_000_000) begin // 5s @ 50 MHz
28                        state <= R;
29                        count <= 29'd0;
30                    end else count <= count + 29'd1;
31                end
32                R: begin
33                    if (count == 29'd500_000_000) begin // 10s @ 50 MHz
34                        state <= G;
35                        count <= 29'd0;
36                    end else count <= count + 29'd1;
37                end
38                default: begin
39                    state <= G;
40                    count <= 29'd0;
41                end
42            endcase
43        end
44    end
45
46    // Combinational: output mapping
47    always @(*) begin
48        case (state)
49            G: \output = 5'b10001;
50            Y: \output = 5'b01001;
51            R: \output = 5'b00110;
52            default: \output = 5'b10001;
53        endcase
54    end
55
56 endmodule
```

The Verilog code in Listing ?? implements the FSM from Section 1. Key design choices made are:

- **Clock division.** The DE1-SoC provides a 50 MHz clock, but the TLC requires delays on the order of seconds. Rather than a separate clock divider module, a 29-bit counter (count) is used directly.
- **State encoding.** The three states are encoded with 2 bits (G=0, Y=1, R=2). A default case catches the unused encoding (3) and returns to state G.
- **Active-low inputs.** The push-buttons output 0 when pressed. The reset is handled via negedge reset in the sensitivity list for asynchronous behaviour, and request is checked against 1'b0.
- **Output mapping.** A combinational always @(*) block maps each state to a 5-bit output vector [4:0] = {veh_G, veh_Y, veh_R, ped_G, ped_R}.

3 Compilation Results

The design was compiled in Quartus Prime 25.1 Lite Edition targeting the Cyclone V 5CSEMA5F31C6 device. The full Compiler Flow Summary screenshot is in Appendix ??; key figures are in Table ??.

Table 2: Compiler Flow Summary.

Resource	Count
Logic utilisation (ALMs)	27
Total registers	32
Total pins	8

The 32 registers break down as follows: the 2-bit state register, the 29-bit count register, and 1 synchroniser flip-flop for the request input. The 27 Adaptive Logic Modules (ALMs) house the counter increment and state transition logic. The 8 pins match the 3 inputs (clk, request, reset) and 5 outputs (output[4:0]).

4 Pin Assignment

During initial compilation, Quartus is free to assign design signals to any FPGA pin. However, the DE1-SoC board has hardwired connections between specific FPGA pins and on-board components (LEDs, push-buttons, clock oscillator), so pin assignments must be set manually via the Assignment Editor to match. The assignments used are shown in Table ?. The design was recompiled after assigning pins so I/O is fitted correctly.

One limitation of the DE1-SoC is that all ten on-board user LEDs (LEDR0–LEDR9) are physically red. The traffic light colours (green, yellow, red) are therefore represented by LED position rather than colour: for example, LEDR8 represents vehicle GREEN even though the physical LED is red. Every output was placed at a different even integer position.

Table 3: Pin assignments for the DE1-SoC board.

Signal	Pin	Component
clk	PIN_AF14	50 MHz Clock
request	PIN_AA15	KEY1 (Push-button)
reset	PIN_AA14	KEY0 (Push-button)
output[0]	PIN_V16	LEDR0 (ped RED)
output[1]	PIN_V17	LEDR2 (ped GREEN)
output[2]	PIN_W17	LEDR4 (veh RED)
output[3]	PIN_Y19	LEDR6 (veh YELLOW)
output[4]	PIN_W21	LEDR8 (veh GREEN)

5 Testing

After programming the FPGA via JTAG, the circuit was tested on-board by exercising each transition in Table ?? and verifying the LED outputs. The results are summarised in Table ??.

Table 4: Board testing results.

Initial State	Input Applied	Expected Next State	Pass/Fail
G	request	Y (5 s timer)	Pass
Y	5 s timeout	R (10 s timer)	Pass
R	10 s timeout	G	Pass
Any	reset	G	Pass

All transitions matched the expected behaviour. Pressing KEY1 (request) in state G immediately turned on LEDR6 (vehicle YELLOW); after approximately 5 s, LEDR4 (vehicle RED) and LEDR2 (pedestrian GREEN) turned on; after a further 10 s the system returned to LEDR8 (vehicle GREEN) and LEDR0

(pedestrian RED). Pressing KEY0 (reset) at any point during the sequence returned the circuit to state G immediately, confirming the asynchronous reset. A photo of the board in state G is shown in Figure ??.

A functional simulation testbench (Listing ?? in Appendix ??) was also used to verify the design in Icarus Verilog prior to board programming, with all 13 assertions passing.

6 Improved Circuit

In the basic TLC, a pedestrian who holds the request button continuously causes the FSM to cycle $G \rightarrow Y \rightarrow R \rightarrow G \rightarrow Y \rightarrow \dots$ with no gap, permanently blocking the vehicle lane on RED. To prevent this, a fourth state L (lockout) is added: after completing state R, the FSM enters L which holds vehicle GREEN for 10s while ignoring any request. Only after the lockout expires does the FSM return to G, where requests are accepted again. The updated state diagram is shown in Figure ??.

6.1 Updated State Machine

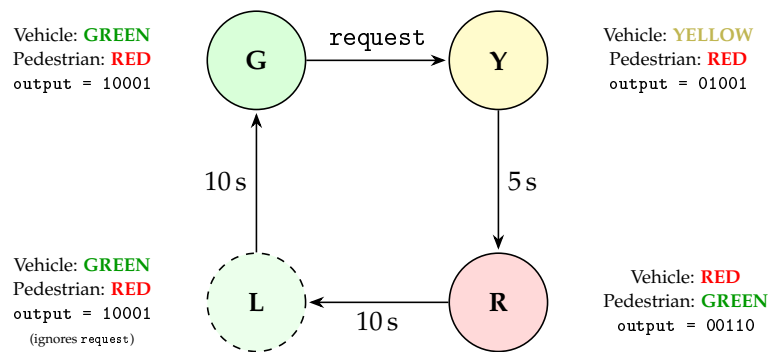


Figure 2: Improved TLC with lockout state L. Output bits as in Figure ??.

After state R, the FSM holds vehicle GREEN for 10 s in L, ignoring requests. reset returns to G from any state.

The key change is that state R now transitions to L (not G), and L transitions to G after a 10s timeout. State L has the same output as G (10001: vehicle GREEN, pedestrian RED) but does not respond to request. All four states fit in the existing 2-bit encoding ($L=3$), so no additional register bits are needed.

6.2 Verilog Implementation

Listing 2: Improved TLC with lockout.

```

1 // Improved Traffic Light Controller with lockout
2 // Adds state L: 10s vehicle GREEN lockout after pedestrian crossing
3
4 module tlc_lock (
5     input wire      clk,
6     input wire      request,
7     input wire      reset,
8     output reg [4:0] \output // {veh_G, veh_Y, veh_R, ped_G, ped_R}
9 );
10
11 localparam G = 2'd0, Y = 2'd1, R = 2'd2, L = 2'd3;
12
13 reg [1:0] state;
14 reg [28:0] count;
15
16 // Sequential: state transitions & timer
17 always @(posedge clk or negedge reset) begin
18     if (!reset) begin
19         state <= G;
20         count <= 29'd0;
21     end else begin
22         case (state)
23             G: begin
24                 if (request == 1'b0) begin
25                     state <= Y;
26                     count <= 29'd0;

```

```

27         end
28     end
29     Y: begin
30         if (count == 29'd250_000_000) begin // 5s @ 50 MHz
31             state <= R;
32             count <= 29'd0;
33         end else count <= count + 29'd1;
34     end
35     R: begin
36         if (count == 29'd500_000_000) begin // 10s @ 50 MHz
37             state <= L;
38             count <= 29'd0;
39         end else count <= count + 29'd1;
40     end
41     L: begin // 10s lockout, ignores request
42         if (count == 29'd500_000_000) begin // 10s @ 50 MHz
43             state <= G;
44             count <= 29'd0;
45         end else count <= count + 29'd1;
46     end
47 endcase
48 end
49 end
50
51 // Combinational: output mapping
52 always @(*) begin
53     case (state)
54         G:      \output = 5'b10001;
55         Y:      \output = 5'b01001;
56         R:      \output = 5'b00110;
57         L:      \output = 5'b10001; // same as G but ignores request
58     endcase
59 end
60
61 endmodule

```

Compared to the basic design, the changes in Listing ?? are minimal: state R transitions to L instead of G, and a new L case reuses the same 10s counter threshold. Since all four 2-bit encodings are now used, the default case is no longer needed.

6.3 Testing

The same test cases from previously were repeated for the improved design, all passing. An additional test verified the lockout: request was held low throughout state L, and the FSM correctly remained in L until the 10s timeout expired before returning to G. The improved circuit was also compiled, programmed onto the board via JTAG, and tested on-hardware with the same results.

7 Conclusions

Both the basic and improved TLC designs were implemented on the Cyclone V FPGA, with all state transitions and timing verified on-board. The lockout state prevented continuous requests from blocking the vehicle lane, and fitted within the existing 2-bit encoding. Verilog behavioural entry made the improvement straightforward compared to a schematic redesign. Overall, the FPGA platform proved well-suited for rapid prototyping of digital control logic: design changes required only re-writing a few lines of Verilog and recompiling, with no hardware modifications. This flexibility, combined with the low resource usage (27 ALMs for the full controller), makes FPGAs a practical choice for deploying custom timing-critical controllers.

A Appendix

A.1 Compilation Report

Flow Summary	
<<Filter>>	
Flow Status	Successful - Fri Feb 20 13:21:29 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	tlc
Top-level Entity Name	tlc
Family	Cyclone V
Device	5CSEMA5F31C6
Timing Models	Final
Logic utilization (in ALMs)	27 / 32,070 (< 1 %)
Total registers	32
Total pins	8 / 457 (2 %)
Total virtual pins	0
Total block memory bits	0 / 4,065,280 (0 %)
Total DSP Blocks	0 / 87 (0 %)
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0 / 6 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 3: Quartus Prime Compiler Flow Summary for the basic TLC design.

A.2 Board Photo

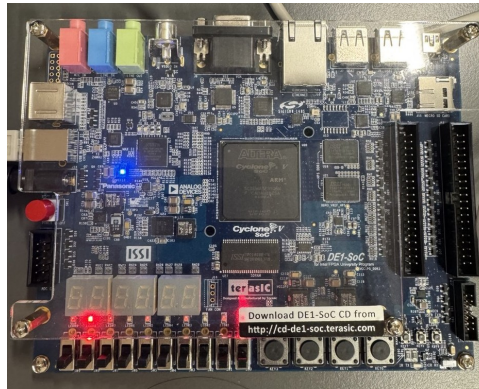


Figure 4: DE1-SoC board in state G: LEDR8 (vehicle GREEN) and LEDR0 (pedestrian RED) are lit.

A.3 Testbench

Listing 3: Testbench Verilog code for the basic TLC.

```
1 // Testbench for basic Traffic Light Controller (tlc.v)
2 // Uses force/release on internal counter to skip long timer waits.
3 `timescale 1ns / 1ps
4
5 module tlc_tb;
6
7     reg clk, request, reset;
8     wire [4:0] out;
9
10    tlc uut (.clk(clk), .request(request), .reset(reset), .\output (out));
11
12    initial clk = 0;
13    always #10 clk = ~clk; // 50 MHz
14
15    localparam G = 2'd0, Y = 2'd1, R = 2'd2;
16    localparam OUT_G = 5'b10001, OUT_Y = 5'b01001, OUT_R = 5'b00110;
17    localparam Y_THRESH = 29'd250_000_000, R_THRESH = 29'd500_000_000;
18
19    integer pass_count = 0, fail_count = 0;
20
21    task check(input [63:0] name, input cond);
22        begin
23            if (cond) begin $display("PASS: %0s", name); pass_count = pass_count + 1; end
24            else begin $display("FAIL: %0s", name); fail_count = fail_count + 1; end
25        end
26    endtask
```

```

27
28 task wait_clks(input integer n);
29     integer i;
30     begin for (i = 0; i < n; i = i + 1) @(posedge clk); end
31 endtask
32
33 // Skip counter to near threshold, then let it trigger naturally
34 task skip_counter(input [28:0] thresh);
35     begin
36         force uut.count = thresh - 2;
37         wait_clks(1);
38         release uut.count;
39         wait_clks(5);
40     end
41 endtask
42
43 initial begin
44     $dumpfile("tlc_tb.vcd");
45     $dumpvars(0, tlc_tb);
46     request = 1; reset = 1;
47
48     // T1: Reset -> state G
49     $display("\n---T1: Reset---");
50     reset = 0; wait_clks(3);
51     check("state=G", uut.state == G);
52     check("out=10001", out == OUT_G);
53     reset = 1; wait_clks(2);
54
55     // T2: Full cycle G -> Y -> R -> G
56     $display("\n---T2: Full cycle---");
57     request = 0; wait_clks(2); request = 1;
58     check("G->Y", uut.state == Y);
59     check("out=01001", out == OUT_Y);
60     skip_counter(Y_THRESH);
61     check("Y->R", uut.state == R);
62     check("out=00110", out == OUT_R);
63     skip_counter(R_THRESH);
64     check("R->G", uut.state == G);
65     check("out=10001", out == OUT_G);
66
67     // T3: Async reset from Y and R
68     $display("\n---T3: Async reset---");
69     request = 0; wait_clks(2); request = 1;
70     check("in_Y", uut.state == Y);
71     reset = 0; wait_clks(2);
72     check("Y_reset->G", uut.state == G);
73     reset = 1; wait_clks(2);
74     // Now reset from R
75     request = 0; wait_clks(2); request = 1; wait_clks(1);
76     skip_counter(Y_THRESH);
77     check("in_R", uut.state == R);
78     reset = 0; wait_clks(2);
79     check("R_reset->G", uut.state == G);
80     reset = 1; wait_clks(2);
81
82     // T4: Idle stability
83     $display("\n---T4: Idle---");
84     request = 1; wait_clks(100);
85     check("stays_G", uut.state == G);
86
87     // Summary
88     $display("\n==%0d passed, %0d failed==", pass_count, fail_count);
89     if (fail_count == 0) $display("ALL TESTS PASSED");
90     else $display("SOME TESTS FAILED");
91     $finish;
92 end
93
94 endmodule

```